Introduction to Coding theory

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We cover the following topics in this note.

- TBA.

Contents

1 Linear Codes

The fundamental problem of coding theory is to transmit information reliably across a noisy channel. A sender begins with a message, usually represented as a vector

$$\mathbf{x} = (x_1, \ldots, x_k) \in \mathbb{F}_2^k,$$

and applies an *encoder*

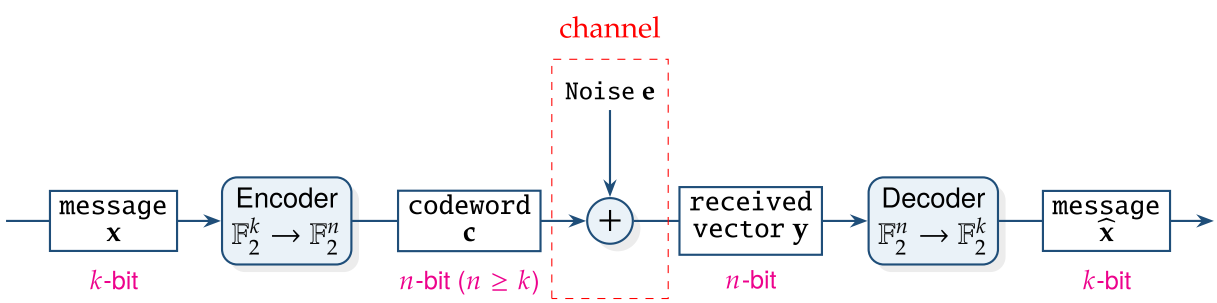
$$\text{Encode} : \mathbb{F}_2^k \longrightarrow \mathbb{F}_2^n, \qquad \mathbf{x} \longmapsto \mathbf{c},$$

where $n \geq k$. The vector $\mathbf{c}$ is called the *codeword*.

During transmission, the codeword passes through a noisy channel. The effect of the channel is modeled by adding an error vector $\mathbf{e} \in \mathbb{F}_2^n$ to the transmitted codeword, so that the receiver obtains

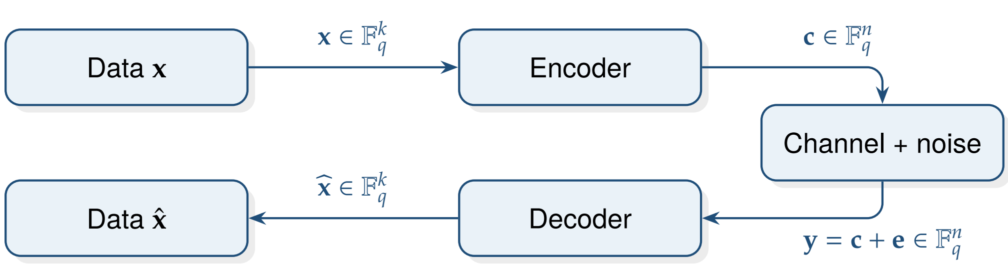
$$\mathbf{y} = \mathbf{c} \oplus \mathbf{e} \in \mathbb{F}_2^n.$$

The vector $\mathbf{y}$ is called the *received vector*. The purpose of the decoder is to recover the original message, or equivalently the transmitted codeword, from the corrupted vector $\mathbf{y}$.



A generic communication model has the form

$$\mathbf{x} = (x_1, \ldots, x_k) \xrightarrow{\text{encoder}} \underbrace{\mathbf{c} = (c_1, \ldots, c_n)}_{\text{codeword } (n \geq k)} \xrightarrow{\text{noise } \mathbf{e}} \mathbf{y} = \mathbf{c} + \mathbf{e} \xrightarrow{\text{decoder}} \widehat{\mathbf{x}} = (\widehat{x}_1, \ldots, \widehat{x}_k).$$



Example 1 (Repetition code). Let Σ be an alphabet. An *encoding* is a map

$$\begin{array}{rccc} \text{Encode} : & \Sigma^k & \longrightarrow & \Sigma^n \\ & (x_1, \dots, x_k) & \longmapsto & (c_1, \dots, c_n) \end{array}$$

with $n \geq k$.

$$x_1 \xmapsto{\text{Encode}} \underbrace{(x_1,\, x_1,\, \dots,\, x_1)}_{n \text{ copies } (\in \Sigma^n)}$$

When $k = 1$, a very simple encoding repeats the same symbol

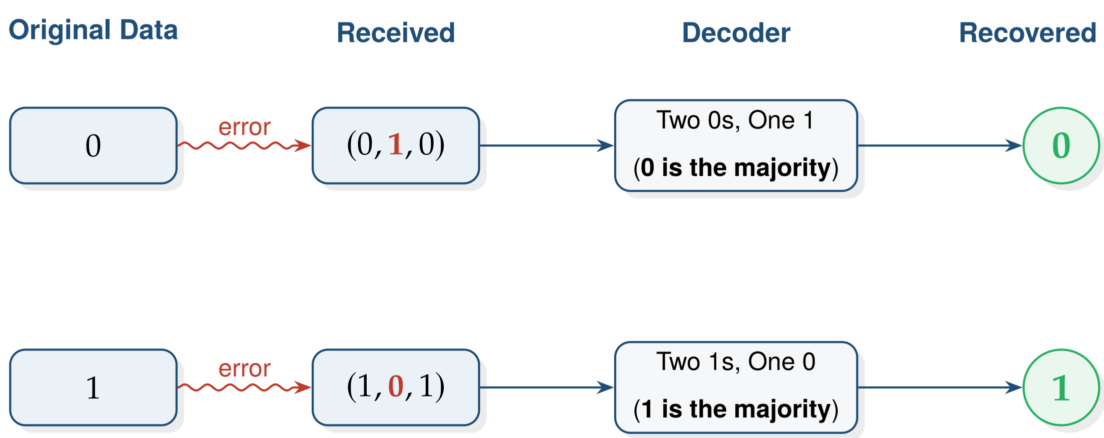
$$0 \longmapsto (0,\, 0,\, \dots,\, 0)$$

$$1 \longmapsto (1,\, 1,\, \dots,\, 1)$$

This is called the *repetition code*. If one symbol is changed during transmission, we can still recover the original message by *majority vote*. For instance,

$$(0, 0, 0) \rightsquigarrow (0, 1, 0), \qquad (1, 1, 1) \rightsquigarrow (1, 0, 1).$$

Since 0 appears most often in $(0, 1, 0)$, we decode it as 0, and since 1 appears most often in $(1, 0, 1)$, we decode it as 1.



1.1 Definition of Linear Codes

Linear Encoding

Definition 1. A **linear encoding** over $\mathbb{F}_q$ is a linear map

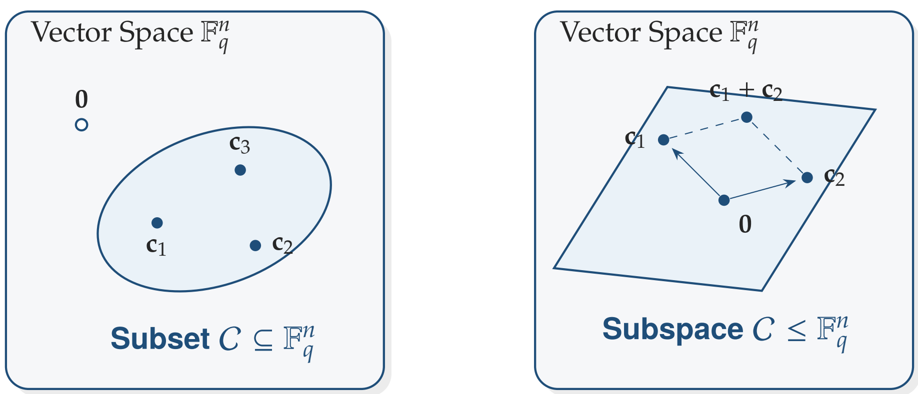
$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear Code

Definition 2. A **code of length** $n \in \mathbb{Z}_{>0}$ **over** $\mathbb{F}_q$ is a subset

$$C \subseteq \mathbb{F}_q^n.$$

The elements of C are called **codewords**. If $C \leq \mathbb{F}_q^n$, then C is called a **linear code**.



Remark 1. If $\dim_{\mathbb{F}_q}(C) = k$, then C is called a $[n,k]_q$**-code over** $\mathbb{F}_q$.

Proposition 1. *If C is an $[n,k]_q$-code, then*

$$|C| = q^k.$$

Proof. Choose a basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ of C. Every codeword can be written uniquely as

$$x_1\mathbf{b}_1 + \cdots + x_k\mathbf{b}_k, \qquad x_i \in \mathbb{F}_q.$$

Each x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k. $\square$

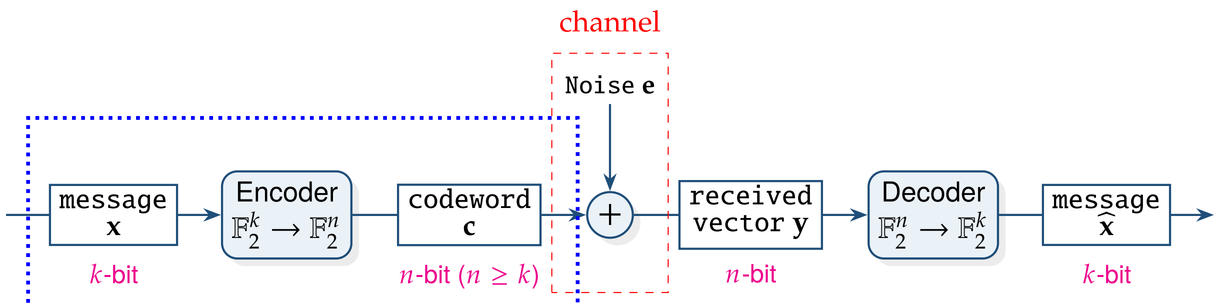
1.2 Generator Matrices

Note. In linear algebra, a choice of bases for finite-dimensional vector spaces V and W identifies each linear transformation $T : V \to W$ with a matrix.

Accordingly, in coding theory, after choosing bases for $\mathbb{F}_q^k$ and $\mathbb{F}_q^n$, a linear encoding

$$\text{Encode} : \mathbb{F}_q^k \to \mathbb{F}_q^n$$

is represented by a matrix $\mathbf{G} \in \mathbb{F}_q^{n\times k}$.

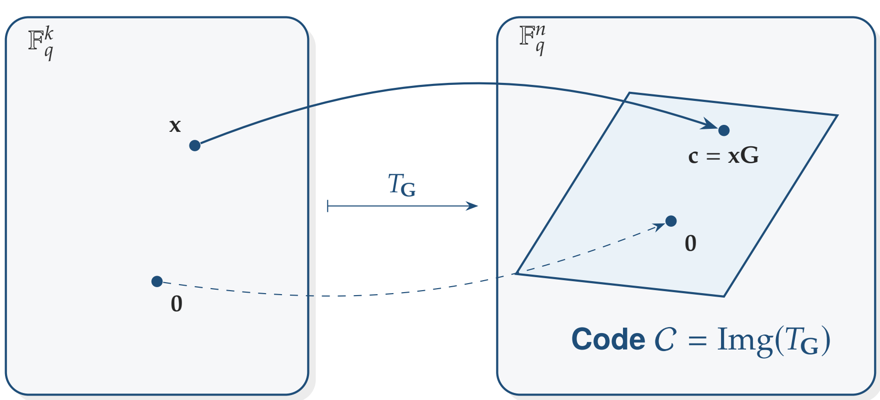


Generator Matrix

Definition 3. Let C be an $[n, k]_q$-code over $\mathbb{F}_q$. A **generator matrix** for C is a $k \times n$ matrix $\mathbf{G}$ whose rows form a basis of C.

Remark 2. If $\mathbf{x} = (x_1, \ldots, x_k) \in \mathbb{F}_q^k$, then encoding is given by

$$\mathbf{c} = \mathbf{x}\mathbf{G} \in \mathbb{F}_q^n \qquad \text{with } \mathbf{G} \in \mathbb{F}_q^{k\times n}.$$

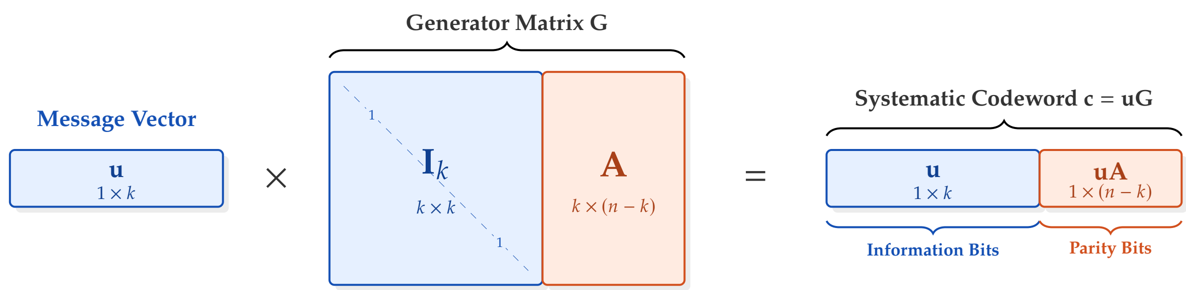


1.2.1 Systematic form

A particularly convenient generator matrix has the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

where $\mathbf{I}_k$ is the $k \times k$ identity matrix and $\mathbf{A}$ is a $k \times (n-k)$ matrix.



Systematic Form

Definition 4. A code is said to be in **systematic form** if it has a generator matrix of the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$$

Remark 3. In general, a linear code has many generator matrices, since different choices of a basis of the code give different matrices. Thus a generator matrix is not unique.

However, if the code is written in *systematic form* with the first k coordinate positions chosen as the information positions, then there is a unique generator matrix of the form $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$.

Lemma 2. *Let $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k\times n}$ be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then there exists a matrix*

$$\exists \mathbf{U} \in GL_k(\mathbb{F}_q) \quad \textit{such that} \quad \mathbf{G}' = \mathbf{U}\mathbf{G}.$$

Proof. Let

$$\mathbf{G} = \begin{pmatrix} — & \mathbf{g}_1 & — \\ — & \mathbf{g}_2 & — \\ & \vdots & \\ — & \mathbf{g}_k & — \end{pmatrix}, \qquad \mathbf{G}' = \begin{pmatrix} — & \mathbf{g}_1' & — \\ — & \mathbf{g}_2' & — \\ & \vdots & \\ — & \mathbf{g}_k' & — \end{pmatrix} \in \mathbb{F}_q^{k\times n}$$

be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then their rows are two bases of the same k-dimensional subspace $C \leq \mathbb{F}_q^n$, so each row of $\mathbf{G}'$ is a linear combination of the rows of $\mathbf{G}$. Thus, there exist scalars $u_{ij} \in \mathbb{F}_q$ such that

$$\begin{aligned}
\mathbf{g}_1' &= u_{11}\mathbf{g}_1 + u_{12}\mathbf{g}_2 + \cdots + u_{1k}\mathbf{g}_k,\\
\mathbf{g}_2' &= u_{21}\mathbf{g}_1 + u_{22}\mathbf{g}_2 + \cdots + u_{2k}\mathbf{g}_k,\\
&\vdots\\
\mathbf{g}_k' &= u_{k1}\mathbf{g}_1 + u_{k2}\mathbf{g}_2 + \cdots + u_{kk}\mathbf{g}_k.
\end{aligned}$$

Let $\mathbf{U} = (u_{ij}) =\in \mathbb{F}_q^{k\times k}$. Then

$$\mathbf{UG} = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ u_{21} & u_{22} & \cdots & u_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix} \begin{pmatrix} - & \mathbf{g}_1 & - \\ - & \mathbf{g}_2 & - \\ & \vdots & \\ - & \mathbf{g}_k & - \end{pmatrix} = \begin{pmatrix} - & (u_{11}\mathbf{g}_1 + \cdots + u_{1k}\mathbf{g}_k) & - \\ - & (u_{21}\mathbf{g}_1 + \cdots + u_{1k}\mathbf{g}_k) & - \\ & \vdots & \\ - & (u_{k1}\mathbf{g}_1 + \cdots + u_{kk}\mathbf{g}_k) & - \end{pmatrix} = \begin{pmatrix} - & \mathbf{g}_1' & - \\ - & \mathbf{g}_2' & - \\ & \vdots & \\ - & \mathbf{g}_k' & - \end{pmatrix} = \mathbf{G}'.$$

So $\mathbf{G}' = \mathbf{UG}$. Similarly, since the rows of $\mathbf{G}$ are also linear combinations of the rows of $\mathbf{G}'$,

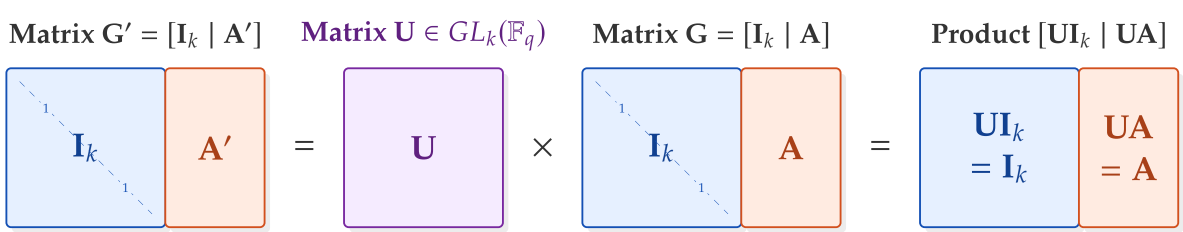
$$\exists \mathbf{V} \in \mathbb{F}_q^{k\times k} \quad \text{such that} \quad \mathbf{G} = \mathbf{VG}'.$$

Since $\mathbf{G}' = \mathbf{UG}$, we have $\mathbf{G} = \mathbf{VUG}$. Therefore $\mathbf{VU} = \mathbf{I}_k$. Similarly, the equation $\mathbf{G}' = \mathbf{UG} = \mathbf{UVG}$ gives $\mathbf{UV} = \mathbf{I}_k$. Therefore $\mathbf{U}$ is invertible, with inverse $\mathbf{V}$. Hence $\mathbf{U}, \mathbf{V} \in GL_k(\mathbb{F}_q)$. □

Note (Uniqueness of Generator Matrices in Systematic Form)**.** If $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$ and $\mathbf{G}' = [\mathbf{I}_k \mid \mathbf{A}']$ are two generator matrices for the same code, then $\mathbf{G}' = \mathbf{UG}$ for some $\mathbf{U} \in GL_k(\mathbb{F}_q)$. Then

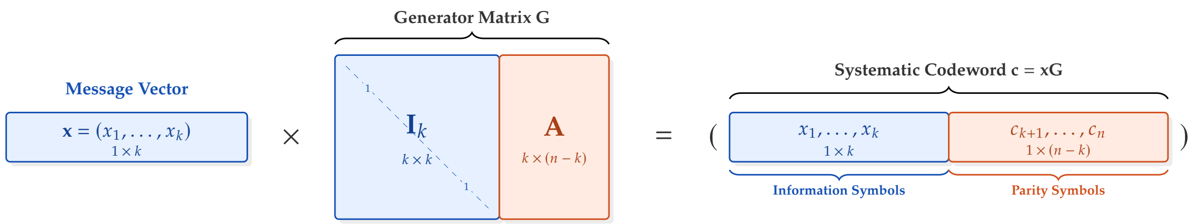
$$\mathbf{G}' = \mathbf{UG} \implies [\mathbf{I}_k \mid \mathbf{A}'] = \mathbf{U}[\mathbf{I}_k \mid \mathbf{A}] \implies [\mathbf{I}_k \mid \mathbf{A}'] = [\mathbf{UI}_k \mid \mathbf{UA}]$$

so $\mathbf{U} = \mathbf{I}_k$. Hence $\mathbf{G}' = \mathbf{G}$, and therefore $\mathbf{A}' = \mathbf{A}$.



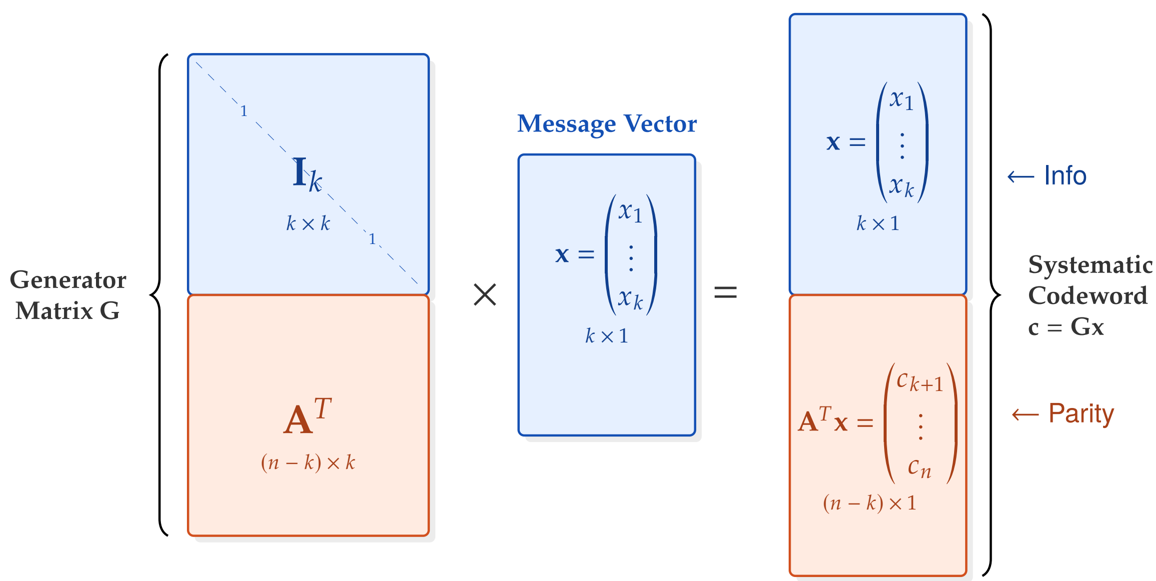
Remark 4. Let the first k coordinates of the codeword are exactly the information symbols. If $\mathbf{x} = (x_1, \ldots, x_k)$, then

$$\mathbf{x}G = (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n).$$



So the information symbols appear in the transmitted codeword, followed by the parity symbols.

Note. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} \in \mathbb{F}_q^k$ be the message vector, written as a column vector. With the column-vector convention, the encoding map is represented by a generator matrix $\mathbf{G}$ such that $\mathbf{Gx} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ c_{k+1} \\ \vdots \\ c_n \end{pmatrix}$.



1.2.2 Examples: Repetition Code and Single Parity-check Code

Example 2 (Repetition code)**.** The $[n,1]_q$ repetition code is the linear code defined by

$$\text{Encode}: \mathbb{F}_q \longrightarrow \mathbb{F}_q^n, \qquad (x_1) \longmapsto (x_1, x_1, \ldots, x_1).$$

Its generator matrix is

$$\mathbf{G} = [1\ 1\ \cdots\ 1]_{1\times n}.$$

Indeed,

$$(x_1)\mathbf{G} = (x_1)[1\ 1\ \cdots\ 1] = (x_1, x_1, \ldots, x_1).$$

Hence the code is

$$C = \{(a, a, \ldots, a) \in \mathbb{F}_q^n : a \in \mathbb{F}_q\} \le \mathbb{F}_q^n.$$

Example 3 (Single Parity-check Code)**.** The single parity-check code $[n, n-1]_q$ is the linear code

$$C = \left\{(x_1, \ldots, x_{n-1}, \textstyle\sum_{i=1}^{n-1} x_i) \in \mathbb{F}_q^n : (x_1, \ldots, x_k) \in \mathbb{F}_q^k\right\} \le \mathbb{F}_q^n.$$

Equivalently, it is defined by the encoding map

$$\text{Encode}: \mathbb{F}_q^{n-1} \longrightarrow \mathbb{F}_q^n, \qquad (x_1, \ldots, x_{n-1}) \longmapsto (x_1, \ldots, x_k, \textstyle\sum_{i=1}^{n-1} x_i).$$

A generator matrix in systematic form is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 1 \end{bmatrix}_{k\times n} = \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = [\mathbf{I}_k \mid \mathbf{1}],$$

where $\mathbf{1} \in \mathbb{F}_q^k$ denotes the column vector all of whose entries are 1. Indeed, for $\mathbf{x} = (x_1, \ldots, x_k) \in \mathbb{F}_q^k$,

$$\mathbf{xG} = (x_1, \ldots, x_k) \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = (x_1, \ldots, x_k, x_1 + \cdots + x_k).$$

Hence the first $n-1$ coordinates are the information symbols, and the last coordinate is their parity symbol.

1.3 Parity-check Matrices

A linear code can be described either by its generators or by the linear equations that all codewords satisfy.

Parity-check Matrix

Definition 5. Let C be an $[n, k]_q$-code. A matrix $H \in \mathbb{F}_q^{(n-k)\times n}$ is called a **parity-check matrix** for C if

$$C = \{\mathbf{v} \in \mathbb{F}_q^n : H\mathbf{v}^\mathsf{T} = 0\}.$$

Thus codewords are exactly the vectors satisfying a system of $n-k$ homogeneous linear equations.

1.3.1 Parity-check Matrix from a Systematic Generator Matrix

Note. Suppose

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

is a generator matrix in systematic form, where $\mathbf{A} \in \mathbb{F}_q^{k\times(n-k)}$. Then define

$$\mathbf{H} = [-\mathbf{A}^\mathsf{T} \mid \mathbf{I}_{n-k}].$$

Parity-check Matrix of a Systematic Generator Matrix

Proposition 3. *Let C be an $[n, k]_q$-code, and suppose that C is generated by the systematic generator matrix*

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}] \in \mathbb{F}_q^{k\times n},$$

where $\mathbf{A} \in \mathbb{F}_q^{k\times(n-k)}$. Then

$$\mathbf{H} = [-\mathbf{A}^\mathsf{T} \mid \mathbf{I}_{n-k}] \in \mathbb{F}_q^{(n-k)\times n}$$

is a parity-check matrix of C.

Proposition 4 (Parity-check matrix of a systematic generator matrix)**.** *Let C be an $[n, k]_q$-code, and suppose that C is generated by the systematic generator matrix*

$$G = [I_k \mid A] \in M_{k\times n}(\mathbb{F}_q),$$

where $A \in M_{k\times(n-k)}(\mathbb{F}_q)$. Then

$$H = [-A^\mathsf{T} \mid I_{n-k}] \in M_{(n-k)\times n}(\mathbb{F}_q)$$

is a parity-check matrix of C.

Proof. By definition, it suffices to show that

$$GH^{\intercal} = 0.$$

Now

$$H^{\intercal} = \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix},$$

so

$$GH^{\intercal} = [I_k \mid A] \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix} = I_k(-A) + AI_{n-k} = -A + A = 0.$$

□

Note (Linear Code as Short Exact Sequence). A linear $[n,k]_q$ code is most naturally described by a short exact sequence

$$0 \longrightarrow \mathbb{F}_q^k \xrightarrow{\text{Encode}} \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0,$$

A message $\mathbf{x} \in \mathbb{F}_q^k$ is encoded as

$$\mathbf{c} = \mathsf{Enc}(\mathbf{x}) \in C.$$

After transmission through the channel, one receives

$$\mathbf{y} = \mathbf{c} + \mathbf{e} \in \mathbb{F}_q^n,$$

where $\mathbf{e} \in \mathbb{F}_q^n$ is the error vector. Applying the parity-check map yields

$$H(\mathbf{y}) = H(\mathbf{c} + \mathbf{e}) = H(\mathbf{e}),$$

since $H(\mathbf{c}) = 0$ for every $\mathbf{c} \in C$.
Thus decoding may be viewed as follows: from the syndrome $H(\mathbf{y})$, choose an error estimate $\widehat{\mathbf{e}}$ with the same syndrome, typically of smallest Hamming weight, and define

$$\widehat{\mathbf{c}} = \mathbf{y} - \widehat{\mathbf{e}} \in C.$$

Then recover the message by

$$\widehat{\mathbf{x}} = \mathsf{Enc}^{-1}(\widehat{\mathbf{c}}).$$

Hence error correction is the procedure of moving the received word from its ambient coset in $\mathbb{F}_q^n/C$ back to the code $C = \ker(H)$, and then inverting the encoding map.

Cardinality of the matrix space $\mathrm{Mat}_m(\mathbb{F}_q)$

Proposition 5. *Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of $m \times n$ matrices over the finite field $\mathbb{F}_q$ is*

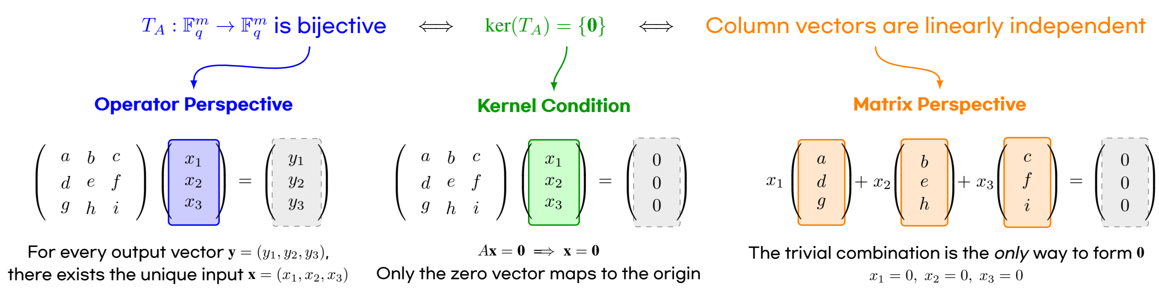
$$\left|\mathrm{Mat}_{m\times n}(\mathbb{F}_q)\right| = q^{mn}.$$

In particular, $\left|\mathrm{Mat}_m(\mathbb{F}_q)\right| = q^{m^2}$.

Remark 5. Let

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_m \\ | & | & & | \end{pmatrix} \in \mathrm{Mat}_{m\times m}(\mathbb{F}_q),$$

where $\mathbf{v}_i \in \mathbb{F}_q^m$ are the columns of A. Then A is invertible if and only if its columns are linearly independent.



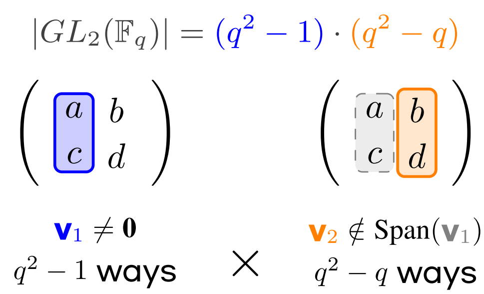
Observation (Computation of $|GL_2(\mathbb{F}_q)|$).

A 2×2 matrix has the form

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad a, b, c, d \in \mathbb{F}_q.$$

Write the columns as

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$



(i) The vector $\mathbf{v}_1$ can be any nonzero vector in $\mathbb{F}_q^2$. Hence there are $q^2 - 1$ choices.

(ii) Since $\mathrm{span}(\mathbf{v}_1) = \{\lambda \mathbf{v}_1 : \lambda \in \mathbb{F}_q\}$ has exactly q elements, there are $q^2 - q$ choices for $\mathbf{v}_2$.

Multiplying these choices, we obtain

$$|GL_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q).$$

Observation (Computation of $|GL_3(\mathbb{F}_q)|$).

$$|GL_3(\mathbb{F}_q)| = (q^3-1)\cdot(q^3-q)\cdot(q^3-q^2)$$

$$\begin{array}{ccccc}
\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} & & \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} & & \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \\
\mathbf{v}_1 \neq \mathbf{0} & & \mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1) & & \mathbf{v}_3 \notin \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2) \\
q^3-1 \text{ ways} & \times & q^3-q \text{ ways} & \times & q^3-q^2 \text{ ways}
\end{array}$$

Observation (Computation of $|GL_4(\mathbb{F}_q)|$).

$$|GL_4(\mathbb{F}_q)| = (q^4-1)\cdot(q^4-q)\cdot(q^4-q^2)\cdot(q^4-q^3)$$

$$\begin{array}{ccccccc}
\begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} & & \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} & & \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} & & \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \\
\mathbf{v}_1 \neq \mathbf{0} & & \mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1) & & \mathbf{v}_3 \notin \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2) & & \mathbf{v}_4 \notin \mathrm{Span}(\mathbf{v}_i)_{i=1}^{3} \\
q^4-1 \text{ ways} & \times & q^4-q \text{ ways} & \times & q^4-q^2 \text{ ways} & \times & q^4-q^3 \text{ ways}
\end{array}$$

Counting invertible matrices over a finite field $\mathbb{F}_q$

Theorem 6. *Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of invertible $m \times m$ matrices over $\mathbb{F}_q$ is*

$$|\mathrm{GL}_m(\mathbb{F}_q)| = (q^m-1)(q^m-q)(q^m-q^2)\cdots(q^m-q^{m-1}).$$

Equivalently, $|\mathrm{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m-q^i)$.

Remark 6. Recall that

$$|\mathrm{Mat}_m(\mathbb{F}_q)| = q^{m^2}, \quad \text{and} \quad |GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m-q^i),$$

and hence

$$\begin{aligned}
\frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \frac{\prod_{i=0}^{m-1}(q^m - q^i)}{q^{m^2}} &= \frac{1}{q^{m^2}}\left[(q^m-1)(q^m-q)(q^m-q^2)\cdots(q^m-q^{m-1})\right] \\
&= \frac{1}{q^{m^2}}\left[q^m\left(1-\frac{1}{q^m}\right)\left(1-\frac{1}{q^{m-1}}\right)\left(1-\frac{1}{q^{m-2}}\right)\cdots\left(1-\frac{1}{q^1}\right)\right] \\
&= \left(1-\frac{1}{q^m}\right)\left(1-\frac{1}{q^{m-1}}\right)\left(1-\frac{1}{q^{m-2}}\right)\cdots\left(1-\frac{1}{q^1}\right) \\
&= \prod_{j=1}^{m}(1-q^{-j}).
\end{aligned}$$

m	$\lvert\operatorname{Mat}_m(\mathbb{F}_q)\rvert$	$\lvert GL_m(\mathbb{F}_q)\rvert$	$\frac{\lvert GL_m(\mathbb{F}_q)\rvert}{\lvert\operatorname{Mat}_m(\mathbb{F}_q)\rvert}$
1	q	$q-1$	$1-q^{-1}$
2	q^4	$(q^2-1)(q^2-q)$	$(1-q^{-1})(1-q^{-2})$
3	q^9	$(q^3-1)(q^3-q)(q^3-q^2)$	$(1-q^{-1})(1-q^{-2})(1-q^{-3})$
4	q^{16}	$(q^4-1)(q^4-q)(q^4-q^2)(q^4-q^3)$	$(1-q^{-1})(1-q^{-2})(1-q^{-3})(1-q^{-4})$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	q^{m^2}	$\prod_{i=0}^{m-1}(q^m-q^i)$	$\prod_{j=1}^{m}(1-q^{-j})$

Example 4 (Invertible binary matrices)**.** For $q = 2$, we have

$$|\operatorname{Mat}_m(\mathbb{F}_2)| = 2^{m^2}, \qquad \frac{|GL_m(\mathbb{F}_2)|}{|\operatorname{Mat}_m(\mathbb{F}_2)|} = \prod_{j=1}^{m}(1-2^{-j}).$$

m	$\lvert\operatorname{Mat}_m(\mathbb{F}_2)\rvert$	$\lvert GL_m(\mathbb{F}_2)\rvert$	$\frac{\lvert GL_m(\mathbb{F}_2)\rvert}{\lvert\operatorname{Mat}_m(\mathbb{F}_2)\rvert}$
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65,536	20,160	0.307617
$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	–	–	0.288788

```
q = 2

print(f"{'m':>2} {'|Mat_m(F_q)|':>15} {'|GL_m(F_q)|':>15} {'ratio':>12}")
print("-" * 48)

for m in range(1, 7):
    Mat = q^(m^2)
    GL = prod(q^m - q^i for i in range(m))
    ratio = GL / Mat
    print(f"{m:>2} {Mat:>15} {GL:>15} {ratio.n():>12.6f}")
```

```
m     |Mat_m(F_q)|      |GL_m(F_q)|         ratio
------------------------------------------------
1               2                1      0.500000
2              16                6      0.375000
3             512              168      0.328125
4           65536            20160      0.307617
5        33554432          9999360      0.298004
6     68719476736      20158709760      0.293348
```

Proposition 7. *Let $q \geq 2$ be fixed, and define*

$$a_m := \frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j}).$$

Then the sequence $\langle a_m \rangle_{m \geq 1}$ converges as $m \longrightarrow \infty$. In other words, there exists $\lim\limits_{n\to\infty} \prod_{j=1}^{m}(1-q^{-j}) \in \mathbb{R}$.

Proof. For every $j \geq 1$,

$$q \geq 2 \implies q^j \geq 2^j > 1 \implies 0 < \frac{1}{q^j} < 1.$$

Therefore

$$0 < \frac{1}{q^j} < 1 \implies -1 < -\frac{1}{q^j} < 0 \implies 0 < 1 - \frac{1}{q^j} < 1.$$

It follows that for each $m \geq 1$,

$$a_m = \prod_{j=1}^{m}(1 - q^{-j}) > 0.$$

So the sequence $\langle a_m \rangle$ is bounded below by 0. Moreover,

$$a_{m+1} = \prod_{j=1}^{m+1}(1-q^{-j}) = \left[\prod_{j=1}^{m}(1-q^{-j})\right] \cdot (1-q^{-(m+1)}) = a_m(1-q^{-(m+1)}).$$

Since $0 < 1 - q^{-(m+1)} < 1$, we obtain $a_{m+1} < a_m$. Thus $\langle a_m \rangle$ is decreasing. Therefore $\langle a_m \rangle$ is a monotone decreasing sequence which is bounded below. Hence it converges. □

Example 5. For small prime q, the limit is approximately

$$q = 2: \qquad \prod_{j=1}^{\infty}(1-2^{-j}) \approx 0.288788095$$

$$q = 3: \qquad \prod_{j=1}^{\infty}(1-3^{-j}) \approx 0.560126077$$

$$q = 5: \qquad \prod_{j=1}^{\infty}(1-5^{-j}) \approx 0.760332795$$

```
q = 2
r = 3
s = 5
R = RealField(100)

L = prod(R(1) - R(q)^(-j) for j in range(1, 100))
M = prod(R(1) - R(r)^(-j) for j in range(1, 100))
N = prod(R(1) - R(s)^(-j) for j in range(1, 100))

print(L)
print(M)
print(N)
```

```
0.28878809508660242127889972193
0.56012607792794894496979224331
0.76033279587123242010148829629
```

1.4 Weights and Distances

A Exercises

References

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